



EETS 7304

Module 5 Part 2 Voice over IP, QoS, Video Conferencing

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Topics/Objectives

- Voice over IP / QoS
- Video Conferencing



Voice over IP

VoIP – troubleshooting, applications, operations and effects of latency and jitter.



Troubleshooting Scenarios

- IP phones might become out of sync in terms of digital certificate verification if network services are not available, are misconfigured, or are simply not reachable.
- An IP phone might not obtain the right amount of power, if CDP is missing.
- A misconfigured DHCP server might prevent IP phones from obtaining
- their configuration files if option 150 is not enabled.
- QoS architectures might render voice traffic useless.
- Security controls might interfere with control protocols and could also filter required signaling protocols, crucial in VoIP operations.
- Protocols and ports in standard IP telephony deployments include:
 - Real-Time Transport Protocol (RTP) and its UDP port ranges
 - Session Initiation Protocol (SIP) on TCP port 5060
 - H323 on TCP port 1720.



Modular QoS CLI (MQC)

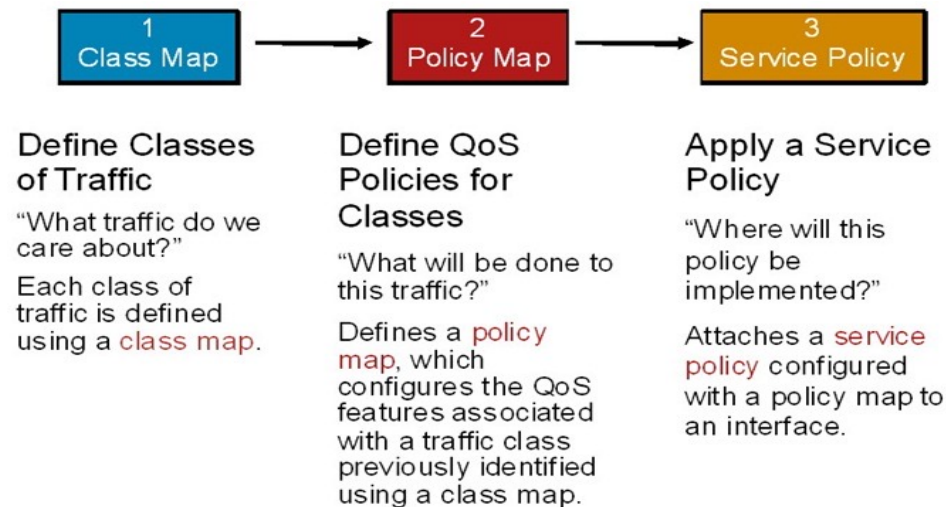
- On most Cisco IOS devices, MQC is used to configure QoS.
- MQC allows you to configure policies once and apply them to multiple interfaces and different devices.
- MQC syntax is not platform specific.
- It decouples the traffic classification components from the policy components.
- You can apply the same policy to different traffic classes without having to create it multiple times.



QoS Policy

Configuring a QoS policy using Cisco IOS MQC has three main components:

- Class maps: Create classification templates for use in policy maps.
- Policy maps: Create a traffic policy to configure the QoS features to be associated with classified traffic.
- Service policy: Assigns a policy map to an interface for incoming or outgoing traffic.



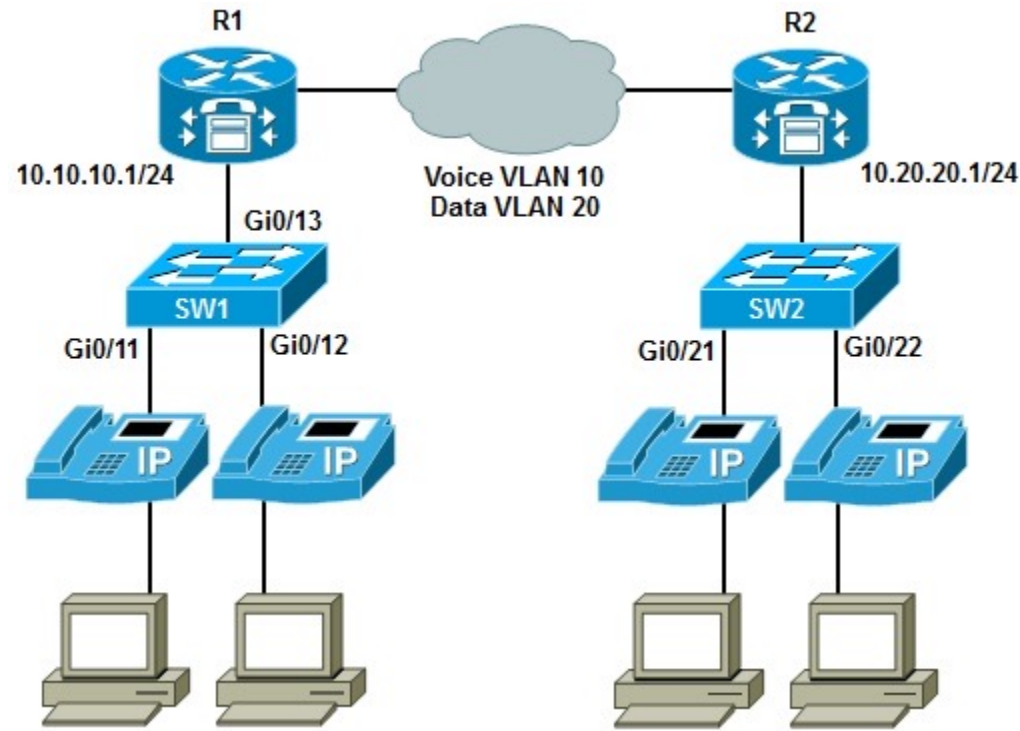
Unified Communications Commands

- After class maps, policy maps, and service policies are configured on the device interfaces, several useful troubleshooting commands are available.
- To summarize the status of the QoS components, use:
 - **show policy-map interface** on routers
 - **show mls qos** on switches
- You can also use appropriate show and debug commands to examine the more traditional services such as DHCP and CDP.
- In converged networks, troubleshoot IP phone issues in this order:
 1. **PoE**
 2. **CDP**
 3. **DHCP**
 4. **TFTP**



Invalid Marking of VoIP Packets

Users from one building complain about their experience with voice calls and claim that it is choppy, they lose connections frequently, and at some point voice conversations are intermittent. the problem is worse for branch-to-branch calls.



Voice Troubleshooting Example – Cont.

- To gathering measurable information, ask the following questions:
 - How often do you observe the reported symptoms?
 - Is there a particular time of the day in which they commonly occur?
 - Is the perceived quality the same when calling internal extension numbers and as it is when calling outside numbers?
 - How often are you unable to obtain a dial tone? For how long does this condition remain?
 - Which locations of the network are experiencing the problem (building/branch)?
 - Are the problematic devices connected to the same wiring closet?
- A comparison to baseline QoS metrics shows that end-to-end delay for voice traffic has doubled across the campus.



Voice Troubleshooting Example – Cont.

- The documentation indicates that IP phones represent the trust boundary, and that the DSCP markings are being used throughout the network.
- Phones are allowed to tag their own packets with high priorities, in this instance DSCP value EF (Expedited Forwarding).
- The command **show mls qos interface** on one of the access switch ports pointing to the phones reveals that the port is trusted and that DSCP values are being maintained and not reset.

```
SW1# show mls qos int g0/11
GigabitEthernet0/11
trust state: trust dscp
trust mode: trust dscp
trust enabled flag: ena
COS override: dis
Default COS: 0
DSCP Mutation Map: Default DSCP Mutation Map
Trust device: none
qos mode: port-based
```



Voice Troubleshooting Example – Cont.

The distribution layer in this network is collapsed at the branch router level. Verify QoS settings on R1. The **show policy-map interface** command reveals that policy “Reclassify” Is Applied to Fa0/0 inbound.

```
R1# show policy-map interface
FastEthernet0/0
Service-policy input: reclassify
Class-map: signaling (match-any)
  0 packets, 0 bytes
  5 minute offered rate 0 bps, drop rate 0 bps
Match: protocol h323
  0 packets, 0 bytes
  5 minute rate 0 bps
Match: protocol sip
  0 packets, 0 bytes
  5 minute rate 0 bps
Match: protocol mgcp
  0 packets, 0 bytes
  5 minute rate 0 bps
QoS set
  dscp af11
  Packets marked 0

Class-map: voice (match-all)
  0 packets, 0 bytes
  5 minute offered rate 0 bps, drop rate 0 bps
Match: protocol rtp audio
  QoS Set
  dscp af31
  Packets marked 0
<output omitted>
```



Voice Troubleshooting Example – Cont.

- The **reclassify** policy attached to the R1 Fa0/0 interface reclassifies and re-marks packets coming into this interface.
- The “QoS Set” section within the VOICE class tells us that VOICE traffic is being classified and tagged with the DSCP value AF31.
- Voice traffic is typically classified with DSCP value EF, the highest priority.
- The voice traffic class is being reclassified into a lower priority and is being incorrectly marked down.
- The impact of this improper remarking is that QoS policies such as bandwidth reservation, priority queuing, and preferred path selection are not enforced.
- Voice traffic is suffering because of the identified voice remarking mistake.
- **This can also be confirmed by examining a Wireshark capture during a voice transmission between the branches.**
 - **Wireshark capture showing markings reclassified.**
- Once this error is fixed, the VOICE problems are solved.
 - Resolve by allowing the values assigned by the IP phones to be trusted across the network.



Video Conferencing.

Identifying use cases, and common issue troubleshooting

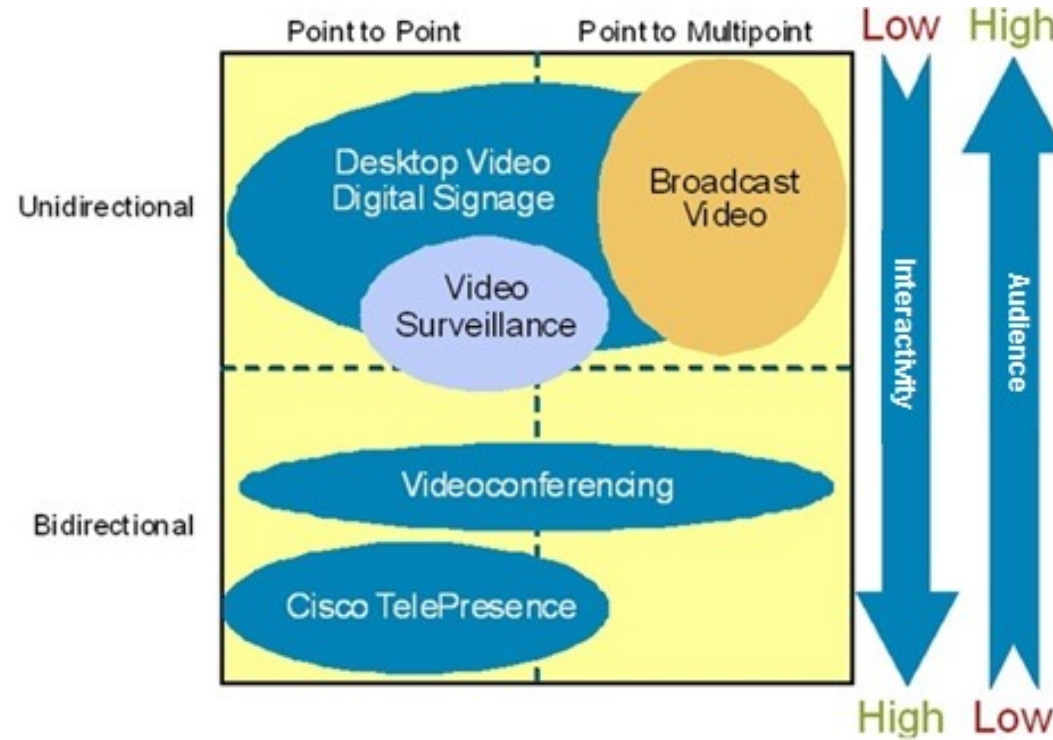


Video Conferencing

- This section addresses the challenge of troubleshooting the network infrastructure supporting video and rich media traffic.
- Several media-rich applications are available for enterprises:
 - High-definition room-based interactive video such as Cisco TelePresence
 - Standard-definition desktop collaboration applications such as Cisco Unified Videoconferencing Systems.
- Streaming and broadcast types of video applications include:
 - Digital signage
 - Video on demand (VoD)
 - Video surveillance.
- Video applications have different characteristics in terms of:
 - Interactivity
 - Network traffic volume
 - Audience
 - Requirements for underlying network infrastructure and services.

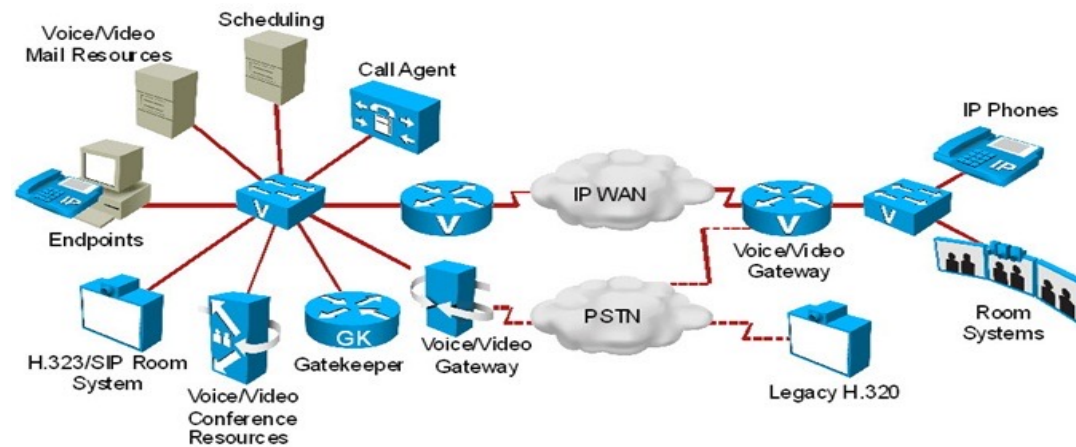


Troubleshooting Video Issues in a Converged Network: Video Application Types



Common Video-Integration Issues

- Several components and infrastructure services are shared between video and voice applications.
- Sometimes the endpoints are the same, or at least integrated and some of the critical protocols, such as SIP, are also the same.
- SIP is a signaling protocol that is used to initiate, manage, and terminate voice calls but also video sessions.
- The end user experiences an integrated service.
- Devices supporting video services in a campus are shown here.



Common Video-Integration Issues 2

- Both Video and voice applications need end-to-end QoS.
- Video is much more bandwidth intensive and very bursty.
- A high-definition stream can require more than 20-Mbps bandwidth for delivery over the network.
- Video packet sizes are much larger.
- Each type of video application has unique requirements and characteristics.
- The table shows the QoS requirements for some of the main video applications.

Metric	Video Collaboration	Cisco TelePresence	Video Surveillance
Latency	200 ms	150 ms	500 ms
Jitter	10 ms	10 ms	10 ms
Loss	0.05%	0.05%	0.5%



Common Video-Integration Issues – Cont.

- Video applications require high availability and millisecond-level network service recovery.
- Video traffic cannot accept unpredictable or large network recovery timeouts.
- Convergence targets will be higher, and packet loss due to network outage must be minimal.
- Redundancy design, convergence of routing protocols and spanning tree are extremely critical.
- Building a multicast-aware network is another important consideration.
- Security in a video-enabled network, similar to voice deployments, might need to permit protocols such as:
 - SIP
 - H.323
 - SCCP (Skinny)
 - RTP
 - RTCP
 - Possibly others



Summary/Takeaways

- Describe basic operation of QoS (i.e. role of DSCP) and how it is used for real-time traffic such as voice and video.
- Characterize the impact of latency and jitter on a network with Voice over IP traffic.
- Apply deployment best practices to create a network topology that will support video conferencing.

